Python Engineering - Class 02

Conditionals and Special Cases | EX02-EX04 - Engineering Decisions

Class objective: Learn to represent decisions, mathematical domains and physical cases with if, elif and else.

This document is the base exercise guide. It is not a full solution to copy: it organizes the problem, implementation steps, reference values and required validation.

Path

```
01_python_bootcamp/
`-- week_01_logic/class_02_conditionals/
    |-- notes/
    `-- exercises/
```

Required workflow

```
PROBLEM
INPUTS
OUTPUTS
EQUATIONS
SPECIAL CASES
DO I NEED ITERATION?
ALGORITHM
VALIDATION
```

Before coding, write these eight sections. The goal is to separate problem reasoning from syntax.

EX02 - Ideal Rocket Equation

Problem: Compute Delta_v using Isp, g0 and the mass ratio.

Reference inputs: Isp, g0, m0, mf

Equations / relationship: Delta_v = Isp * g0 * ln(m0/mf)

Expected check: Use a valid reference such as Isp=250 s, g0=9.81, m0=500000 kg, mf=250000 kg -> Delta_v about 1699.0 m/s.

Implementation guide

- 1. Validate Isp and masses.
- 2. Check m0 > mf.
- 3. Compute mass_ratio.
- 4. Apply np.log().
- 5. Display Delta_v.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

EX03 - Quadratic Equation Solver

Problem: Solve ax^2 + bx + c = 0 while handling linear, repeated, real and complex-root cases.

Reference inputs: a, b, c

Equations / relationship: D = b**2 - 4*a*c; x = (-b +/- sqrt(D))/(2*a)

Expected check: Test: (1,-3,2)->1,2; (1,2,1)->-1; (1,0,1)->complex; a=0 -> linear case.

Implementation guide

- 1. Validate real coefficients.

- 2. Handle a=0 before the quadratic formula.
- 3. Compute discriminant.
- 4. Separate D<0, D=0 and D>0.
- 5. Substitute roots back into the equation.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

EX04 - Piecewise Drag Coefficient

Problem: Compute Mach from speed and temperature and select C_D from three regions.

Reference inputs: speed=500 m/s; T=20 degC; gamma=1.4; R=287

Equations / relationship: a = sqrt(gamma*R*T_K); M=speed/a; Cd=0.3 if M<0.8, 0.5 if M<1.2, else 0.25

Expected check: At 20 degC, speed of sound is about 343 m/s, so M about 1.46 and C_D=0.25.

Implementation guide

- 1. Validate speed and temperature.
- 2. Convert degC to K.
- 3. Compute speed of sound.
- 4. Compute Mach.
- 5. Select C_D with if/elif/else.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

Final validation

Answer these questions as comments at the end of the exercise. The goal is not memorized wording but demonstrating that you can explain the code flow.

- Why does if/elif ordering matter?
- What is the difference between independent if statements and an if/elif/else chain?
- What does the discriminant represent?
- Why must a=0 be handled before the quadratic formula?
- Why does elif M < 1.2 already represent 0.8 <= M < 1.2 in EX04?

Completion criterion

The class is complete when the code runs, the reference checks make sense, you can explain every block without reading a solution, and you can modify at least one input while predicting the expected effect.